College of Engineering – Department of Civil Engineering

CURRICULUM

Bachelor of Science in Geological Engineering

Academic Year 2025-2026

Reference CMOs: CMO No. 35 S 2008, CMO No. 20 s. 2013, CMO No. 39 s. 2021, CMO No. 40 s. 2021

Curriculum Description

The Bachelor of Science in Geological Engineering is a four-year undergraduate program that integrates geology, engineering principles, and field-based learning. Students will study core subjects such as mineralogy, petrology, structural geology, geomorphology, rock and soil mechanics, hydrogeology, geotechnical engineering, exploration techniques, and environmental geology.

The curriculum combines theoretical and practical approaches through lectures, laboratory work, field mapping, modeling, data interpretation, and design projects. Students will also be exposed to Geographic Information Systems (GIS), remote sensing, drilling technology, and resource estimation techniques. The program emphasizes safety, ethics, environmental stewardship, and community engagement in addressing real-world problems.

Program Education Objectives

The geological engineering alumni three to five years after graduation shall:

PEO 1: Specialist.
Practice as specialist in solving complex geological engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.

PEO 2: Professionalism and Leadership.
Assume leadership position in industry, academe, government, or private sector within consideration to social and ethical responsibility.
PEO 3: Lifelong Learning.

Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:

SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex geological engineering problems;

SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for

public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO5 Problem Analysis. Identify, formulate, and solve complex geological engineering problems by applying principles of engineering, science, and mathematics;

SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

SO7 Communication. Communicate effectively on complex geological engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex geological engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

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CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	4	0	4
Integral Calculus	4	0	4
Probability and Statistics for AI	3	0	3
Differential Equations	3	0	3
Subtotal	**14**	**0**	**14**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers 1	3	3	4
Physics for Engineers 2	2	3	3
Subtotal	**8**	**9**	**11**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer-Aided Design	0	3	1
Engineering Economics	3	0	3
Technopreneurship	3	0	3
Engineering Mechanics	3	0	3
Fundamentals of Deformable Bodies	3	0	3
Fluid Mechanics	3	0	3
Basic Occupational Safety and Health	3	0	3
Methods of Research	3	0	3
OJT/Internship (320 hrs)	320 hrs		4
Subtotal	**21**	**9**	**28**
D. Allied Courses			
Analytical Chemistry	3	3	4
General Surveying 1	2	3	3
Geotechnical Engineering: Soil Mechanics	3	3	4
Petroleum Geology	2	3	3
Computer Programming	0	3	1
Subtotal	**10**	**15**	**15**
E. Professional Courses			
Geological Engineering Orientation	1	0	1
Principles of Geology	2	3	3
Elementary and Optical Mineralogy	3	3	4
Paleontology and Historical Geology	3	0	3
Advanced Engineering Mathematics for GeoE	3	0	3
Petrology and Petrography	3	3	4
Structural Geology and Tectonics	3	3	4
Geology of the Philippines and Southeast Asia	3	0	3
Engineering Geomechanics and Rock Stability	2	3	3
Geochemistry	2	3	3

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Geomorphology	2	3	3
Fundamentals of Geographic Information System and Remote Sensing	2	3	3
Principles of Geophysics	2	3	3
Environmental Geology and Engineering	2	3	3
Economic Geology	3	0	3
Stratigraphy	2	3	3
GeoEng Design Project 1	0	3	1
Geological Engineering Field Methods	0	3	1
Resource Exploration and Geostatistics	3	0	3
Foundation Engineering	3	0	3
Geohazard Assessment and Analysis	3	0	3
Drilling Engineering	3	0	3
Seminars and Colloquia in Geological Engineering	0	3	1
Hydrogeology and Groundwater Engineering	3	0	3
Volcanology and Geothermal Systems	3	0	3
Tunneling and Excavation Engineering	3	0	3
Geology, Environment, and Mining Laws, Energy Policies, and Ethics	3	0	3
Geological Engineering Project Management	3	0	3
GeoEng Design Project 2	0	3	1
Subtotal	**65**	**45**	**80**
TOTAL TECHNICAL COURSES	**118**	**78**	**148**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics in the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology, and Society	3	0	3
Subtotal	**24**	**0**	**24**
B. General Education Elective and Mandated Courses			
Linear Algebra for AI	3	0	3
Machine Learning and Neural Networks	3	0	3
The Entrepreneurial Mind	3	0	3
Life and Works of Rizal	3	0	3
Subtotal	**12**	**0**	**12**

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C. Physical Activities Toward Health and Fitness (PATHFit)			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL NON-TECHNICAL COURSES	**50**	**0**	**50**
GRAND TOTAL	**168**	**78**	**198**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	14
B. Natural/Physical Sciences	11
C. Basic Engineering Sciences	28
D. Allied Courses	15
E. Professional Courses	80
II. NON-TECHNICAL COURSES	
A. General Education Courses	24
B. General Education Elective and Mandate Courses	12
C. Physical Activities Toward Health and Fitness (PATHFit)	8
D. National Service Training Program	6
GRAND TOTAL	**198**

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CURRICULUM STRUCTURE

FIRST YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 101	Introduction to Engineering	1	0	3		
GEd 101	Understanding the Self	3	3	0		
GEd 102	Mathematics in the Modern World	3	3	0		
GEd 105	Readings in Philippine History	3	3	0		
GEd 106	Purposive Communication	3	3	0		
MATH 101	Differential Calculus	4	4	0		
NSTP 111	National Service Training Program 1	3	3	0		
PATHFit 1	Movement Competency Training	2	2	0		
CHEM 101	Chemistry for Engineers	4	3	3		
	Total	**26**	**24**	**6**		

FIRST YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
CpE 101	Computer Programming	1	0	3		
ENGG 102	Engineering Drawing	1	0	3		
GEd 104	The Contemporary World	3	3	0		
GEd 108	Art Appreciation	3	3	0		
GEd 109	Science, Technology and Society	3	3	0	ENGG 101	
MATH 102	Integral Calculus	4	4	0	MATH 101	
NSTP 121	National Service Training Program 2	3	3	0	NSTP 111	
PATHFit 2	Exercise-based Fitness Activities	2	2	0	PATHFit 1	
PHYS 102	Physics for Engineers 1	4	3	3	MATH 101	MATH 102
	Total	**24**	**21**	**9**		

FIRST YEAR						
MIDTERM SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
GEd 103	Life and Works of Rizal	3	3	0		
PHYS 103	Physics for Engineers 2	3	2	3	PHYS 102	
GEd 107	Ethics	3	3	0		
	Total	**9**	**8**	**3**		

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SECOND YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
MATH 103	Differential Equations	3	3	0	MATH 102	
ENGG 109	Engineering Mechanics	3	3	0	PHYS 102	
GeoE 201	Geological Engineering Orientations	1	1	0		
GeoE 202	Principles of Geology	3	2	3		
CHEM 201	Analytical Chemistry	4	3	3	CHEM 101	
GE 202	General Surveying 1	3	2	3	ENGG 102	
ENGG 103	Computer-Aided Design	1	0	3	ENGG 102	
GEdE 101	The Entrepreneurial Mind	3	3	0		
AI 101	Linear Algebra for AI	3	3	0		
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 1	2	2	0	PATHFit 1 & 2	
	Total	**26**	**22**	**12**		

SECOND YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
GeoE 203	Elementary and Optical Mineralogy	4	3	3	CHEM 201, GeoE 202	
GeoE 204	Paleontology and Historical Geology	3	3	0	GeoE 202	
GeoE 205	Advanced Engineering Mathematics for GeoE	3	3	0	Math 103	
ENGG 112	Fluid Mechanics	3	3	0	PHYS 103, ENGG 109	
AI 102	Probability and Statistics for AI	3	3	0	AI 101	
ENGG 110	Fundamentals of Deformable Bodies	3	3	0	ENGG 109	
ENGG 114	Basic Occupational Safety and Health	3	3	0	ENGG 101	
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor, and Adventure Activities 2	2	2	0	PATHFit 1 & 2	
	Total	**24**	**23**	**3**		

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THIRD YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
GeoE 301	Petrology and Petrography	4	3	3	GeoE 203	
GeoE 302	Structural Geology and Tectonics	4	3	3	GeoE 203, GeoE 204	
CE 306	Geotechnical Engineering: Soil Mechanics	4	3	3	ENGG 110	
GeoE 303	Geology of the Philippines and Southeast Asia	3	3	0	GeoE 204	
GeoE 304	Engineering Geomechanics and Rock Stability	3	2	3	ENGG 110	
ENGG 116	Methods of Research	3	3	0	AI 102, 3rd year Standing	
AI 103	Machine Learning and Neural Networks	3	3	0	AI 101, AI 102, CpE 101	
	Total	**24**	**20**	**12**		

THIRD YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
GeoE 305	Geochemistry	3	2	3	GeoE 301	
GeoE 306	Geomorphology	3	2	3	GeoE 301, GeoE 302	
GeoE 307	Fundamentals of Geographic Information System and Remote Sensing	3	2	3	GE 202, AI 103	
GeoE 308	Principles of Geophysics	3	2	3	PHYS 103, GeoE 301, GeoE 302	
GeoE 309	Environmental Geology and Engineering	3	2	3	GeoE 301, GeoE 302, CE 306	
GeoE 310	Economic Geology	3	3	0	GeoE 301, GeoE 302	
GeoE 311	Stratigraphy	3	2	3	GeoE 301, GeoE 302, GeoE 303	
ENGG 104	Engineering Economics	3	3	0	3rd year standing	
GeoE 312	GeoEng Design Project 1	1	0	3	ENGG 116, GeoE 301, GeoE 302	
	Total	**25**	**18**	**21**		

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THIRD YEAR						
MIDTERM SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
ENGG 105	Technopreneurship	3	3	0	3rd year standing	
GeoE 313	Geological Engineering Field Methods	1	0	3	GeoE 306, GeoE 307, GeoE 311	
PetE 203	Petroleum Geology	3	2	3	GeoE 308, GeoE 310, GeoE 311	
	Total	**7**	**5**	**6**		

FOURTH YEAR						
FIRST SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
GeoE 401	Resource Exploration and Geostatistics	3	3	0	AI 103, GeoE 305, GeoE 310	
CEG 402	Foundation Engineering	3	3	0	CE 306, GeoE 304	
GeoE 402	Geohazard Assessment and Analysis	3	3	0	GeoE 313, GeoE 309	
PetE 427	Drilling Engineering	3	3	0	CE 306, GeoE 304, GeoE 306	
GeoE 403	Seminars and Colloquia in Geological Engineering	1	0	3	GeoE 312, GeoE 313	
GeoE 404	Hydrogeology and Groundwater Engineering	3	3	0	ENGG 112, CE 306, GeoE 313	
GeoE 405	Volcanology and Geothermal Systems	3	3	0	GeoE 305, GeoE 306, GeoE 308	
GeoE 406	Tunneling and Excavation Engineering	3	3	0	GeoE 304, CE 306, GeoE 306	
GeoE 407	Geology, Environment, and Mining Laws, Energy Policies, and Ethics	3	3	0	4th year standing	
	Total	**25**	**24**	**3**		

FOURTH YEAR						
SECOND SEMESTER						
Course Code	**Course Title**	**Credit Units**	**Lec**	**Lab**	**Prerequisite(s)**	**Co-requisite(s)**
GeoE 409	GeoEng Design Project 2	1	0	3	GeoE 312	
GeoE 408	Geological Engineering Project Management	3	3	0	GeoE 313, GeoE 407	
ENGG 117	OJT/Internship (320 hrs)	4	320		4th year standing	
	Total	**8**	**3**	**3**		